

Drivebase Hardware Map

Project: FRC Java Coding Lab 7.0 - Real Robot Programming

Lesson: D01_L10_Basic_Integrated_Robot

Version: 1.1

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Purpose

This document records the current drivetrain hardware configuration and serves as the single source of truth for the drivebase.

Drivebase Overview

Item	Value
Drive Type	Tank Drive
Motor Controller	REV SPARK MAX
Motor Type	Brushed CIM
Motor Count	4
Power Distribution	REV PDH
Main Controller	RoboRIO 2
Framework	WPILib 2026

Drivebase CAN Mapping

CAN ID	Device	Role	Motor	Notes
10	SPARK MAX	Left Leader	Left CIM	Main left drivetrain controller
7	SPARK MAX	Left Follower	Left CIM	Follows CAN 10
11	SPARK MAX	Right Leader	Right CIM	Main right drivetrain controller
8	SPARK MAX	Right Follower	Right CIM	Follows CAN 11

Leader / Follower

Left: CAN 10 (Leader) -> CAN 7 (Follower)

Right: CAN 11 (Leader) -> CAN 8 (Follower)

Motor Configuration

Setting	Value
Current Limit	60 A
Voltage Compensation	12.0 V
Left Leader Inverted	TRUE
Right Leader Inverted	FALSE
Follower Mode	Enabled

Controller Mapping

Button	Function
Left Stick Y	Left Tank Drive
Right Stick Y	Right Tank Drive
RT	Intake
LT	Outtake
RB	Feed
LB	Reverse Feed
X	Flywheel Only
Y	Full Shooter: Flywheel 0.60 -> 1.0 s Spin-up Delay -> Feeder 0.40
A, B, Start, Back	Reserved

Software Flow

Xbox Controller → RobotContainer → DriveSubsystem → DriveIO → DriveIOSparkMax →
Leaders → Followers → Wheels

Verification

- CAN 10 Left Leader ID VERIFIED
- CAN 7 Left Follower ID VERIFIED
- CAN 11 Right Leader ID VERIFIED
- CAN 8 Right Follower ID VERIFIED
- Clean Build PASS
- Simulation PASS
- Driver Station / Glass PASS
- Real Robot PASS

D01_L10 Hardware Identity Correction

Previous lessons used an incorrect physical Left/Right drivetrain identity. D01_L10 corrects the hardware mapping so software Left controls the physical left drivetrain and software Right controls the physical right drivetrain. No command, subsystem, IO contract, or telemetry architecture was changed.